

ATPL EASA STUDY GUIDE | 022 - AGK INSTRUMENTATION

Chapter 4: Air Data Computers (ADC) & ADIRU Architecture

Syllabus Source: CAE Oxford (pp. 119-148) | Pass Target: >= 90% AviationExam

1. AIR DATA COMPUTER (ADC) INPUTS & COMPUTED OUTPUTS

Modern glass-cockpit airliners replace individual mechanical instrument capsules with centralized digital Air Data Computers (ADC):

ADC Functional Category	Aeronautical Parameters & Connected Systems
Raw Sensor Inputs to ADC	Pitot Pressure (P _t), Static Pressure (P _s), Total Air Temperature (TAT), Angle of Attack (Alpha).
Primary Computed Outputs	Pressure Altitude (1013.25 hPa), Baro-corrected Altitude (QNH/QFE), Calibrated Airspeed (CAS), True Airspeed (TAS), Mach Number, Static Air Temperature (SAT), Vertical Speed (VSI).
Downstream Consumer Systems	EFIS Primary Flight Displays, Autopilot (AFCS), Flight Management System (FMS), TCAS II, GPWS/EGPWS, FADEC engine controllers, and Transponder Mode S Altitude Reporting.

- Electronic Position Error Correction (SSEC): The ADC automatically applies Static Source Error Corrections (SSEC) stored in digital calibration tables based on current Mach number, AOA, and flap position, achieving flawless altitude and speed precision.

2. AIR DATA INERTIAL REFERENCE UNIT (ADIRU) ARCHITECTURE

Modern commercial airliners (e.g. Airbus A320/A350/A380, Boeing 777/787) integrate the ADC and the Inertial Reference System (IRS) into a single unified line replaceable unit called an ADIRU:

- ADIRU Segregation: Each ADIRU contains two distinct, fault-isolated parts: 1. The ADR (Air Data Reference) part, processing pitot/static/temp data; 2. The IR (Inertial Reference) part, processing laser gyro and accelerometer attitude/position data.
- Triple Redundancy: Transport category aircraft install three independent ADIRUs (ADIRU 1, ADIRU 2, ADIRU 3). ADIRU 1 supplies the Captain's PFD/ND; ADIRU 2 supplies the First Officer's PFD/ND; ADIRU 3 acts as a standby hot backup selectable via cockpit switching.
- ADIRU Rotary Mode Selector: OFF: Unit depowered; NAV: Normal operating mode (inertial navigation active); ATT: Attitude-only reversion mode (used if navigation position computation fails in flight; provides basic pitch, roll, and heading, but velocity/position navigation is lost!).

[!] EXAM TRAP (AVIATIONEXAM) - ADIRU Switching & In-Flight Reversion
- If an ADR fault occurs: The pilot switches the Air Data selector to CAPT ON 3 or F/O ON 3. - If an IR fault occurs in flight: The pilot switches the mode selector to ATT (Attitude mode). The aircraft MUST be flown straight and level for 30 seconds to level the attitude platform. Entering magnetic heading manually via MCDU is mandatory in ATT mode!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Air Data Switching Protocol

Scenario: Unreliable airspeed alert on Captain PFD due to pitot icing on Probe

- Step 1: Cross-check Captain PFD, First Officer PFD, and Standby ISFD. Step 2: On overhead AIR DATA switching panel, turn Captain selector from 'NORM' to 'CAPT ON 3'. Step 3: Captain PFD now receives valid air data from standby ADIRU 3, clearing the flag.

3. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: Which of the following parameters is directly COMPUTED by an Air Data Computer (ADC) rather than sensed?

- [A]: Pitot pressure
- [B]: Total Air Temperature
- [C]: True Airspeed (TAS) and Mach Number
- [D]: Angle of attack

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [C]. Pitot pressure, static pressure, TAT, and AOA are raw physical inputs. True Airspeed, Mach number, and calibrated airspeed are mathematically computed by the ADC processors.

Q2: In an aircraft equipped with three ADIRUs, what capability remains if an Inertial Reference (IR) unit is switched off?

- [A]: Full LNAV and RNAV navigation
- [B]: Pitch, roll attitude, and heading information ONLY (navigation capability is lost)
- [C]: Autoland CAT III capability
- [D]: TCAS resolution advisories

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The ATT (Attitude) mode is an emergency reversion. It retains basic gyro attitude (pitch/roll) and magnetic heading, but completely disables inertial position, groundspeed, and navigation tracking.

Q3: What is the operational purpose of Static Source Error Correction (SSEC) within an Air Data Computer?

- [A]: To de-ice the pitot probe
- [B]: To automatically compensate for aerodynamic position errors around the static vents across various Mach numbers and angles of attack
- [C]: To calibrate cockpit compasses
- [D]: To measure fuel density

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Static vents suffer position error caused by airflow pressure disturbances. The ADC applies SSEC algorithms to eliminate position error, ensuring highly accurate altitude and airspeed readouts.